WORKED SOLUTION

STEP I 1988, Question 14

Probability and Statistics • Censored Normal Expectation

STEP I 1988, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s1-q14

The key idea

The variable X_M is just $\min(X, M)$: the normal variable with everything above M squashed down to M. Its expectation splits into an integral over $x < M$ and a constant times a tail probability, and the integral is easy because $x\phi(x)$ is, up to sign, the derivative of $\phi(x)$. The tourist story is the same idea in disguise. The number of passengers a bus fleet carries is $\min(\text{demand}, \text{capacity})$, and after a normal approximation that is exactly an X_M.

Solution

The expectation of X_M

Since $X_M = X$ on $\{X < M\}$ and $X_M = M$ on $\{X \geq M\}$,

$$\mathrm{E}(X_M) = \int_{-\infty}^{M} x\,\phi(x)\,\mathrm{d}x \;+\; M\,\mathrm{P}(X \geq M) = \int_{-\infty}^{M} x\,\phi(x)\,\mathrm{d}x \;+\; M(1-\Phi(M)).$$

? **Ask yourself** Is there a function whose derivative is $x\phi(x)$?

Yes. With $\phi(x) = \frac{1}{\sqrt{2\pi}}\mathrm{e}^{-x^2/2}$, the chain rule gives $\phi'(x) = -x\phi(x)$. So

$$\int_{-\infty}^{M} x\,\phi(x)\,\mathrm{d}x = \Big[-\phi(x)\Big]_{-\infty}^{M} = -\phi(M) + 0,$$

because $\phi(x) \to 0$ as $x \to -\infty$. Hence

$$\mathrm{E}(X_M) = -\phi(M) + M(1-\Phi(M)),$$

as required.

As a sanity check, letting $M \to \infty$ gives $\mathrm{E}(X_M) \to 0 = \mathrm{E}(X)$, since $M(1-\Phi(M)) \to 0$. Letting $M \to -\infty$ gives $\mathrm{E}(X_M) \approx M$, as it should.

The buses

Let N be the number of tourists, on one visit, who are sent to the bus station. We assume that each of the 1024 tourists is directed independently, with probability $\frac{1}{2}$ each way. Then

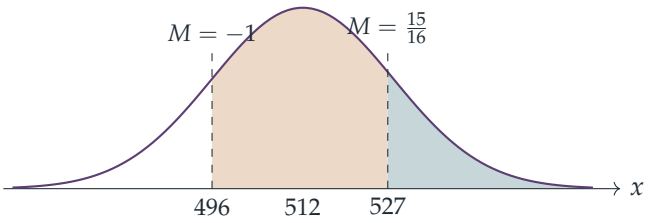
$$N \sim \mathrm{B}(1024, \tfrac{1}{2}), \qquad \mathrm{E}(N) = 512, \qquad \mathrm{Var}(N) = 1024 \cdot \tfrac{1}{2} \cdot \tfrac{1}{2} = 256 = 16^2.$$

With $n = 1024$ this large, N is very well approximated by $512 + 16X$ with X standard normal.

With 16 buses the capacity is $16 \times 31 = 496$. With 17 it is 527. A fleet of capacity C carries $\min(N, C)$ passengers, so its profit for one visit, in vloska, is

$$\begin{aligned} \min(N, C) &\approx \min(512 + 16X,\ C) \\ &= 512 + 16 \min\Big(X, \frac{C - 512}{16}\Big) = 512 + 16\, X_M, \qquad \text{where } M = \frac{C - 512}{16}. \end{aligned}$$

So $C = 496$ gives $M = -1$, and $C = 527$ gives $M = \frac{15}{16}$.



When demand lands in the amber band, the 17th bus carries the excess over 496. When demand lands in the teal tail, it carries a full 31.

Using $\phi(1) \approx 0.24197$ and $\Phi(1) \approx 0.84134$, and noting $1 - \Phi(-1) = \Phi(1)$,

$$\mathrm{E}(X_{-1}) = -\phi(1) - \big(1 - \Phi(-1)\big) = -\phi(1) - \Phi(1) \approx -0.24197 - 0.84134 = -1.08332.$$

Using $\phi(0.9375) \approx 0.25707$ and $1 - \Phi(0.9375) \approx 0.17425$,

$$\mathrm{E}(X_{15/16}) \approx -0.25707 + 0.9375 \times 0.17425 \approx -0.25707 + 0.16336 = -0.09371.$$

The extra bus therefore raises the expected profit per visit by

$$16\big(\mathrm{E}(X_{15/16}) - \mathrm{E}(X_{-1})\big) \approx 16 \times 0.98961 \approx 15.83 \text{ vloska.}$$

There are 50 visits a year, so the extra bus is worth about

$$50 \times 15.83 \approx 792 \text{ vloska a year.}$$

This is the largest annual bribe worth considering. Paying any more than the expected extra profit would leave the cooperative worse off on average.

Ask yourself Shouldn't we use a continuity correction, since N is discrete?

Here it isn't needed. A continuity correction helps when we approximate a probability such as $\mathrm{P}(N \leq c)$, where the function being averaged jumps at an integer. The function $k \mapsto \min(k, C)$ has no jump, and the uncorrected normal approximation is extremely accurate for it. The exact binomial calculation gives 791.68, which agrees with the figure above to two decimal places. Shifting the capacities by $\frac{1}{2}$ would actually make things worse, giving 775.

Checks

- Midpoint quadrature of $\int_{-\infty}^{M} x\phi(x)\,\mathrm{d}x + M(1-\Phi(M))$ agrees with the formula to eight decimal places for several values of M. A Monte Carlo simulation of $\min(X, M)$ with 4×10^5 samples agrees too.
- Exact summation over $\mathrm{B}(1024, \frac{1}{2})$ gives an extra expected profit of 15.8337 per visit, which is 791.68 a year, matching the normal approximation.
- Rough plausibility: the extra bus earns something whenever $N > 496$, which happens with probability about 0.84, and it is often full. So a figure a little over half of 31 per visit is reasonable.

Common slips

- Forgetting to scale: $\min(N, C)$ is $512 + 16X_M$, so the difference in $\mathrm{E}(X_M)$ must be multiplied by the standard deviation 16, and then by 50.
- Sign errors with $M = -1$: the term $M(1-\Phi(M))$ is $-(1-\Phi(-1)) = -\Phi(1)$, not $-(1-\Phi(1))$.

Beyond STEP

The quantity $\mathrm{E}\min(X, M)$ is the expected value of a *censored* normal variable. Closely related expressions, such as $\mathrm{E}(X-M)^+ = \phi(M) - M(1-\Phi(M))$, are the backbone of inventory theory (the "newsvendor problem") and of the Black–Scholes price of an option. The identity $\mathrm{E}\min(N, C+1) - \mathrm{E}\min(N, C) = \mathrm{P}(N > C)$ is a neat way to see the value of each extra unit of capacity. For more on these ideas, see Grimmett and Welsh, *Probability: An Introduction*.

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